



Fabrication and characterization of coconut shell activated carbon using variation chemical activation for wastewater treatment application

E.H. Sujiono^{a,*}, D. Zabrian^a, Zurnansyah^a, Mulyati^a, V. Zharvan^a, Samnur^b, N.A. Humairah^c

^a Laboratory of Materials Physics, Department of Physics, Universitas Negeri Makassar, Makassar 90224, Indonesia

^b Department of Mechanical Engineering, Universitas Negeri Makassar, Makassar 90224, Indonesia

^c Department of Physics, Universitas Sulawesi Barat, Sulawesi Barat 91412, Indonesia

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ABSTRACT

Activated carbon based on coconut shell has been successfully synthesized using three different chemical activators. The coconut shell was obtained from the highland with the freshwater environment and dried by solar thermal. The sample was carbonized at 600°C for 3 h and thoroughly activated using three chemical activating agents: NaOH, H₃PO₄, and ZnCl₂. All samples were then characterized using X-ray Diffraction (XRD), Fourier Transform Infrared (FTIR), Surface Area Analysis (SAA), and Scanning Electron Microscopy-Energy Dispersive X-Ray (SEM-EDX) to identify their crystal structure, functional group, surface area, and morphology. The XRD result illustrates amorphous phase has been formed with carbon graphite as the highest content while hydrogen (H), oxygen (O), nitrogen (N), potassium (K), sodium (Na), phosphorus (P), calcium (Ca) and magnesium (Mg) as the impurities. The FTIR characterization results show no significant change from the three kinds of activators. The presence of -OH, C=O, and C-O functional groups indicates that the activated carbon has been successfully conducted. The SAA characterization gives the highest surface area value (516 m²g⁻¹) obtained by the NaOH activator. The results of morphology identification show a similar pore structure for each sample with less impurity content for different pore distribution confirmed by EDX results. Additionally, the Total Dissolved Solids (TDS) measurement shows the activated carbon with an activating agent of NaOH after filtering the polluted water meets the standard requirements for drinking water. All of those results confirm that the activated carbon-based coconut shells can potentially be applied for water purification treatment.

Introduction

Activated carbon is a widely used material in the chemical industry [1–4]. Superior physicochemical properties that activated carbon have such as high surface area, physically and chemically stable make activated carbon has a wide range of applications, such as water purification, gas filter, supercapacitor electrode, and catalyst support [3,5–8]. Two activation methods can produce activated carbon: physical activation and chemical activation [9,10]. Physical activation is carbon activation with pyrolyzed under an inert atmosphere, while chemical activation is activating process with chemical activator impregnation such as KOH, NaOH, NaCO₃, ZnCl₂, AlCl₃, H₃PO₄, H₂SO₄, and some acids [11–13]. Chemical activation is favored over physical activation because it can work at lower temperatures with less activation time [14–16]. Chemical activation is mainly considered a reaction between the solid precursor and the chemical activator. The concentration,

temperature, and activation time determine the extent of the reaction of organic acids [17]. Chemical activation is beneficial for inhibiting formation, eliminating other volatile organics compounds, and increasing the surface area in activated carbon [5,18–20]. In addition, chemical and physical activation provide different alternatives in fabricating activated carbon to produce adsorbents with other physicochemical characteristics. Activated carbon produced by carbonization and physical activation does not have satisfactory performance as adsorbent. Meanwhile, activated carbons fabricated by chemical activation before or after carbonization have a narrow micropore distribution, high surface area, and high adsorption capacity [21].

The demand for activated carbon increases day by day. Carbon material as the precursor to fabricate activated carbon is truly crucial. The porous structure of activated carbon obtained using the chemical activation method has attracted many researchers because of their high specific surface area. Pore volume, high electrical conductivity, tunable

* Corresponding author.

E-mail address: e.h.sujiono@unm.ac.id (E.H. Sujiono).

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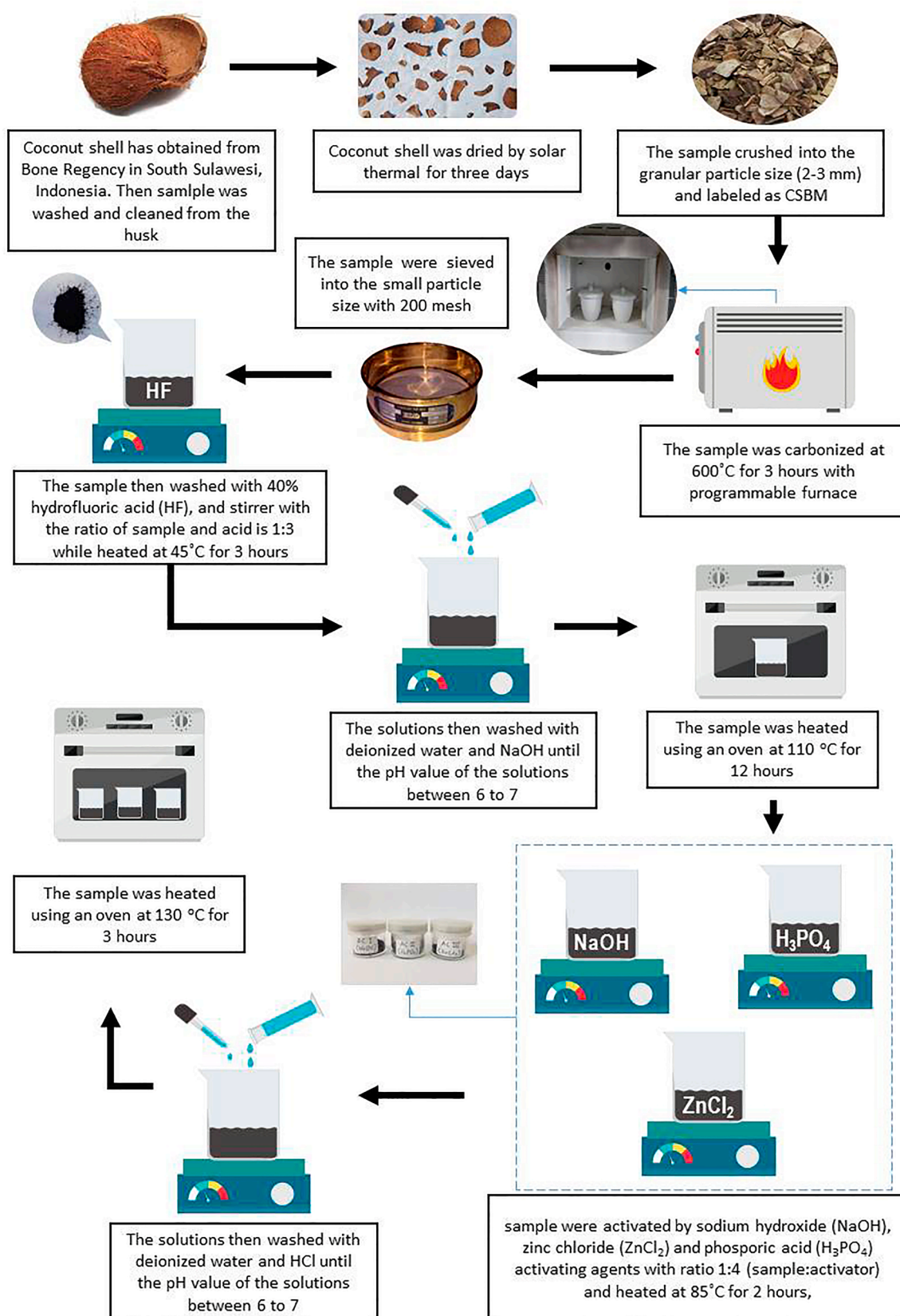


Fig. 1. Schematic of synthesis activated carbon base coconut shells with different activating agents.

porosity, strong chemical, thermal, and mechanical stability, hydrophobic nature of their surfaces, low density, chemical inertness, ease of handling, and low cost, porous carbons have piqued interest. They are important since they are used as industrial adsorbents in air and water purification, pollutant adsorption, gas separation, catalyst supports, templates and electrode materials, chromatography columns, gas capture, and storage [22]. However, the activated carbon produced from carbonaceous material with high carbon and low inorganic content is relatively expensive [23]. This fact has encouraged a growing interest in making low-cost activated carbon [24,25]. Recently, activated carbon-based biomass waste with high carbon has been produced [10]. Besides, activated carbon is cheap, made based on biomass is more preferred because of abundant and easily obtained [9]. Many researchers have intensively worked to produce more qualified activated carbon than more efficient and cheaper biomass wastes [26,27–28]. Producing activated carbon from biomass waste has been reported by several researchers, such as coffee shells, corncob, rice straw, wheat straw, and coconut shells [29].

In many countries, the waste of coconut is not optimally used as a high-value product. The high production of coconut means that the creation of coconut shell wastes is so high. Several downstream industries result from coconut processing, such as cosmochemical, coco fiber, coconut oil, desiccated coconut, and nata de coco [30]. The utilization of coconut shell waste right now applied as charcoal for grilled and souvenirs or merchandise, whereas this coconut shell can be a raw material for producing a valuable product with high economic value such as activated carbon and graphene materials [19,31–33].

The purpose of this work is to utilize the coconut shell waste in South Sulawesi province, Indonesia, by fabricating an activated carbon with different activating agents: sodium hydroxide (NaOH), alkali earth metal salts, zinc chloride (ZnCl_2), and phosphoric acid (H_3PO_4). Toprak et al. (2017) reported that activated carbon using NaOH treatment has high surface areas, micropore volumes, and adsorption capacities [34]. Meanwhile, Vargas et al (2012) obtained that the activation of carbon materials with the chemical activation of H_3PO_4 and ZnCl_2 produces microporous materials that can absorb CO_2 compounds [35]. Therefore, all three activating agents were used in this research.

In this fundamental research, the best activating agent will be used in the measurement quality of polluted water. The influence of activating agents is investigated by using XRD to investigate crystal structures of the sample, SAA to determine the surface area and the pore size distribution, FTIR is used to analyze the functional groups of the samples, SEM-EDX to identify the surface morphology and elemental composition of the samples. TDS measurement measures the number of substances contained in polluted water and after filtering with activated carbon.

Experimental procedure

Fabrication of coconut shell charcoal in powder form

Coconut shells were collected from Bone regency, Indonesia. The sample was washed and cleaned from the husk then dried by solar thermal for three days. The sample was then grinded into the granular size (2 mm – 3 mm). The sample of coconut shell was carbonized at 600°C for 3 h to produce a rich carbon coconut shell charcoal [3]. The sample was grinded and sieved until obtained 200 mesh of particle size. Next, the sample in powder form was washed with 40% hydrofluoric (HF) to remove the impurity compound of the sample [33,36]. The sample was cleaned using a hotplate stirrer by adjusting the sample ratio, and HF was 1:3 and heated at 45°C for 3 h. The sample was then washed with deionized water until the pH value of the solutions between 6 and 7 and dried with an oven at 110 °C for 12 h.

Synthesis of activated carbon

Afterward, the sample was activated by using sodium hydroxide

Table 1

Match! analysis result of coconut shell charcoal after washing with HF.

Formula	Legend	%
Carbon Graphite 2H	C	71.53
Calcium-beta	Ca	10.02
Phosphorus – black	P	14.47
Potassium	K	3.97

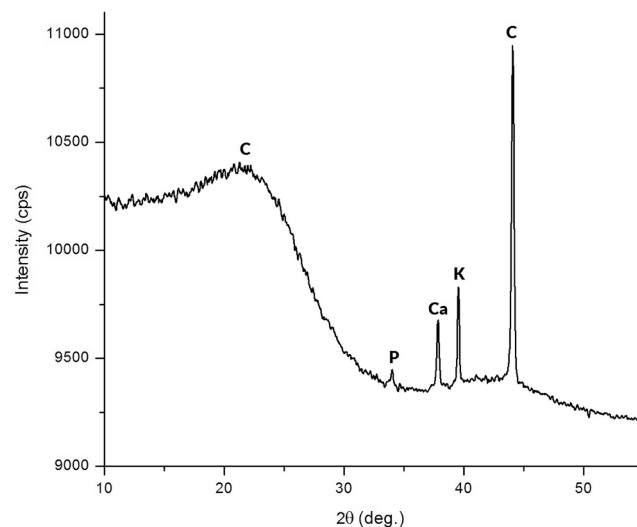


Fig. 2. XRD diffractogram pattern with elemental content in coconut shell charcoal after washing with hydrofluoric acid (HF).

(NaOH), zinc chloride (ZnCl_2), and phosphoric acid (H_3PO_4) activating agents with a ratio between sample and activator is 1:4 and heated at 85°C for 2 h then dried at 130°C for 3 h. After activating, the sample was washed with 1 M hydrochloric acid (HCl) and deionized water until the pH 7 was obtained, then dried with an oven at 110°C for 12 h. All chemical analytic reagents grades were purchased from Merck-Germany. Schematic of synthesis activated carbon based on coconut shells with different activating agents shown in Fig. 1.

Characterization

The XRD analysis was performed using Shimadzu XRD 7000 diffractometer with acceleration voltage was equal to 40 kV and the current 30 mA. Scanning diffractometer recorded at 2θ range 10 to 55. Characterization of SEM-EDX was performed using FEI Inspect S50 to identify the morphology and elemental percentage of the activated carbon. Meanwhile, characterization of FTIR was performed using IR Tacer-100 to identify the transmittance spectrum of the sample and functional groups of the activated carbon with the wavenumber measurement range from 700 to 4000 cm^{-1} . The Quadra Sorb Station 1, Version 7.01, measured the samples surface area, pore size, pore distribution, and surface volume. SAA was determined by N_2 adsorption with 77 K bath temperature.

Results and discussion

Activated carbon based on coconut shell charcoal was successfully produced using three different activating agents. The powder samples were characterized using XRD, SEM-EDX, FTIR, and SAA. The qualified sample was then tested using a TDS meter to measure the effect of an activating agent on the quality of water.

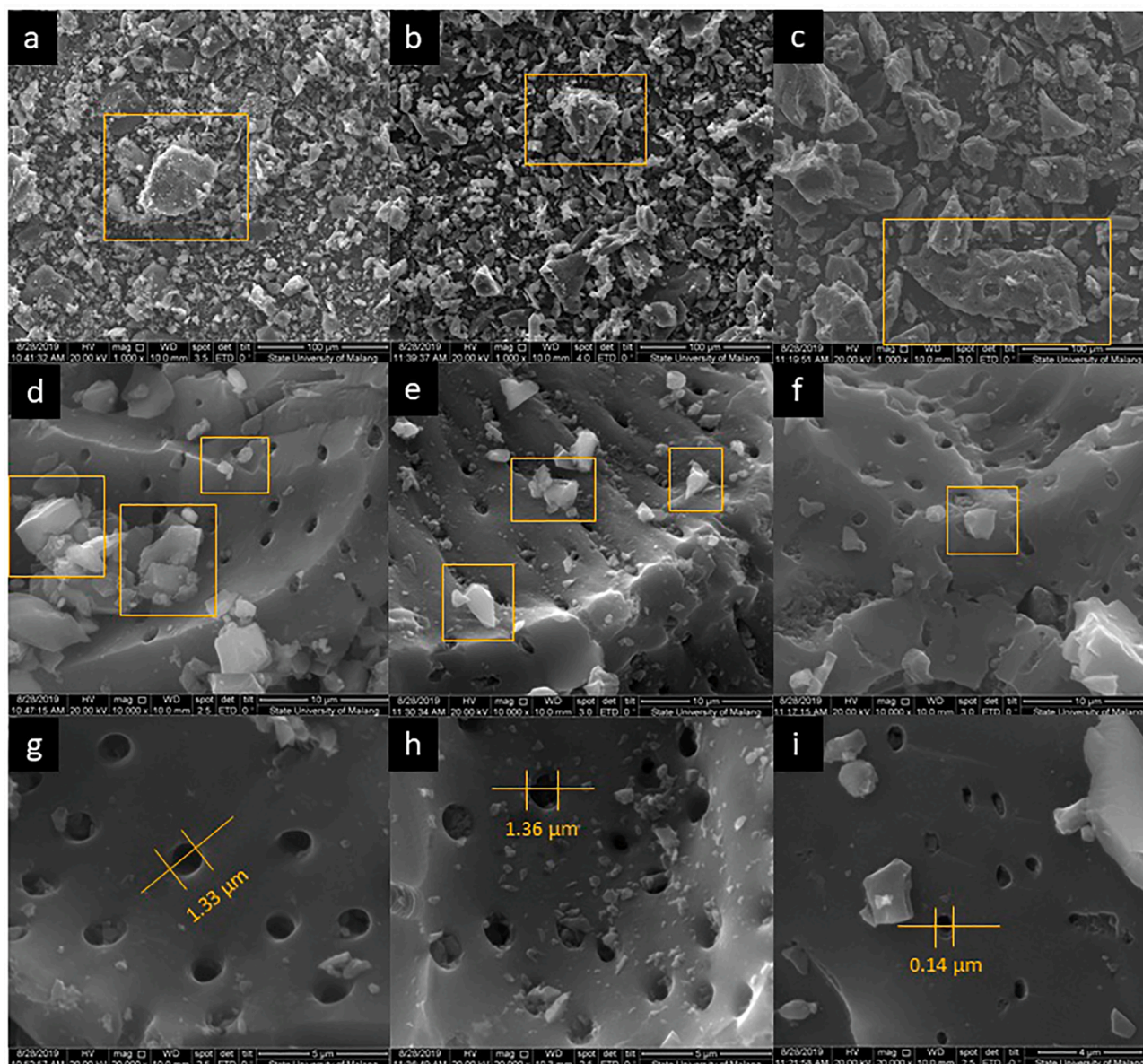
XRD analysis

XRD characterization shows the result of diffractogram patterns of

Table 2

Microstructural XRD data of the three highest peaks of coconut shell charcoal.

No.	Peak no.	Two thetas (deg)	d (Å)	I_n/I_1 (%)	FWHM (deg)	Intensity (Counts)	Integrated Int (Counts)
1	36	44.0462	2.05424	100	0.18530	1312	13,895
2	33	39.5240	2.27823	30	0.15370	395	3505
3	32	37.8129	2.37729	20	0.19240	258	3314

**Fig. 3.** Surface morphology of the sample AC I (a,d,g), AC II (b,e,h), and AC III (c,f,i).

the coconut shell charcoal sample which contains information regarding the elemental content and phases after being washed by hydrofluoric acid (HF). The sample was then analyzed using Match! Software to obtain the sample phase, which the results of this analysis can be seen in Table 1. Based on Table 1, it is crystal clear that the graphitization process was successfully conducted with a high graphite phase occurring in the sample [37]. Fig. 2 shows that the amorphous phase has been formed due to the raw materials used from natural resources. While, other impurities such as hydrogen (H), oxygen (O), nitrogen (N), potassium (K), sodium (Na), phosphorus (P), calcium (Ca), and magnesium (Mg) occurred due to the presence of soil macronutrients which naturally contained in coconut shell [38,39] and nonoptimal of the washing

process.

In Table 2, the microstructure of the sample was gained based on the parameters of 2θ data, intensity, and FWHM (Full-Width at Half Maximum), which were illustrated by the first three highest elemental peaks in a sample of coconut shell charcoal after being washed with 40% hydrofluoric (HF). It can be seen that the first highest peak is Carbon (graphite 2H) with an angle of 2θ at 44.0462°, FWHM value at 0.18530°, and intensity at 1312 cps. The second highest peak is K (potassium) with the angle 2θ at 39.5240°, FWHM value at 0.15370°, and intensity at 395 cps. Then, the third-highest peak is found for Ca (calcium-beta) with the angle of 2θ at 37.8129°, FWHM value at 0.19240°, and intensity at 258 cps. All of those FWHM values give information about the crystal

Table 3

Elemental composition of activated carbon-based coconut shell with different chemical activator by EDX analysis.

Element	AC I (NaOH)	AC II (ZnCl)	AC (H ₃ PO ₄)
C K	90.56	90.44	93.32
O K	07.66	07.45	07.48
Na K	00.93	–	–
Si K	00.30	–	–
P K	00.32	–	00.39
Ca K	00.24	–	–
Zn K	–	00.93	–
Cl K	–	01.18	–

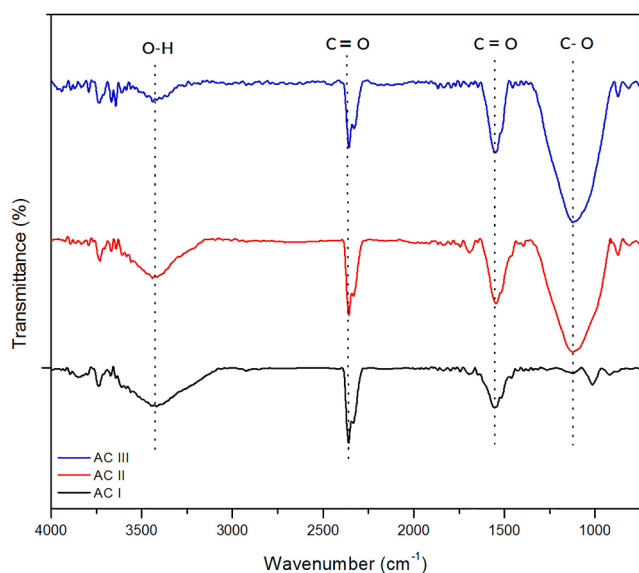


Fig. 4. FTIR spectra of activated carbon-based coconut shell by different chemical activator.

homogeneity of the coconut shell charcoal. A smaller FWHM value corresponds to the sample's homogeneous lattice and crystal structure, corresponding to good materials quality [40,41].

SEM-EDX analysis

SEM characterized the morphological surface of the samples. The characterization of each sample can be seen in Fig. 3 (a,b,c). It can be seen that an image with 1000 times magnification shows the different grain particle sizes of each sample. The AC III sample has a larger grain particle size than the AC I and II. The morphological surface in Fig. 3 (d, e,f) shows the different color degradation of the SEM image, which illustrates AC I and AC II has much grayscale color at the grain particles. It is quite different from AC III, which has a dominant dark at the grain particle. The other color degradation, as indicated by the impurity

content of the sample caused by the organic sample and activation process. The results of the EDX characterization can be seen in Table 3. The EDX result shows that the sample with low grayscale degradation colors such as AC I and AC II has a high purity content than the sample with high grayscale degradation colors, AC III. The image of SEM by 20,000 times magnification can be seen in Fig. 3 (g,h,i). The morphology samples illustrate pore structure in the surface with different pore distributions of each sample [42].

It has been proved that the fabrication of activated carbon has been successfully fabricated. The pore structure of the sample was the essential properties of activated carbon. The analysis result of the surface sample shows AC I, AC II, and AC III have a pore size of about 1.33 μm , 1.36 μm , and 0.14 μm , respectively. The formed pore structure on the surface indicates samples have the macropore size classification or pore size above 50 nm [43]. The surface area analysis proved that the pore structure was on micropore classification and the macropore area.

FTIR analysis

Fig. 4 shows the FTIR pattern with functional groups of the samples. The FTIR characterization used transmittance spectrum data and was recorded in 750–4000 cm^{-1} . The characterization results show that a functional group on a sample surface was close to each other. The FTIR spectra of the AC I with NaOH chemical activator were found at peak 3442.94 cm^{-1} , 2360.87 cm^{-1} , and 1556.55 cm^{-1} . Sample AC II, which used the H₃PO₄ chemical activating agent, showed at peak 3415.93 cm^{-1} , 2357.01 cm^{-1} , 1543.05 cm^{-1} , and 1122.57 cm^{-1} . And the AC III sample, which used the ZnCl₂ as an activating agent, showed at peak 3439.08 cm^{-1} , 2357.01 cm^{-1} , 1552.70 cm^{-1} , and 1120.64 cm^{-1} . The peak at approximately 3400 cm^{-1} was due to O–H vibrations, which indicated the phenolic groups [21,44,45]. The peak at 2350 cm^{-1} and 1550 cm^{-1} corresponded to a carboxylic acid with a stretching vibration of C=O [4,28,46]. And the peak at 1120 cm^{-1} was attributed to C–O [47,48].

The FTIR results also showed a significant difference of the sample at peaks 1120 cm^{-1} where there is no peak at AC I. The functional groups of activated carbon, which obtained O–H, C=O, and C–O stretching vibration, have the essential role as an absorbent of pollutant [49]. Further, the functional groups which contained oxygen on the sample surface are a depositional location to absorb the metal particle on catalyst application [47]. The functional groups containing an activated carbon-based coconut shell have similarities with previous research and a commercial activated carbon [49].

SAA analysis

The SAA is a characterization to identify the surface properties of a porous material, such as surface area, pore size, and pore volume [50]. The properties of porous material influence the application of its material. The surface area is a characteristic possessed by the porous material, which has an essential role in the absorption application [51]. The high surface area shows a high amount of absorption of porous material. This absorption property has many applications in the industry, such as

Table 4

Comparison of the surface area characteristics of activated carbon based on a coconut shell and bio-waste reported by Kopac (2012)

Sample	BET surface area (m ² /g)	Micropore Area (m ² /g)	Micropore volume [V _{mic}] (cm ³ /g)	Pore volume (cm ³ /g)	Pore Width (nm)	Average pore diameter (nm)
AC I	516	391.74	71.53	0.160	1.29	1.79
AC II	42	19.71	0.012	0.023	1.43	3.46
AC III	23	17.97	0.008	0.011	1.19	3.48
Empty fruit bunches (EFB)	687.34	580.23	0.297	0.045	2.11	–
Tamarind seeds	1784	–	0.62	0.93	1.1	–
Corn cob	3708	2359	0.372	1.628	–	–
Rice Husk	16	–	0.003	0.026	–	–

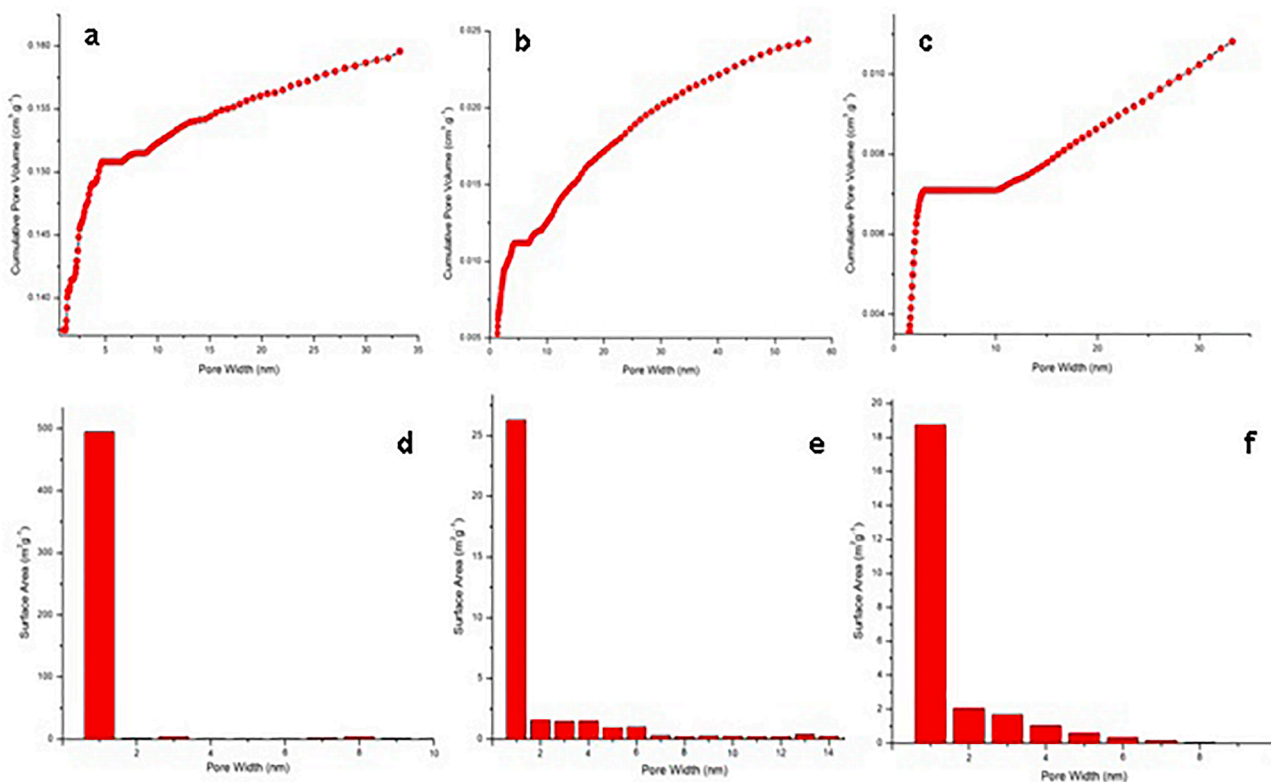


Fig. 5. The graphic pattern of cumulative pore volume with pore width to produce pore volume value of AC I(a), AC II(b), and AC III(c) and the histogram data of pore size distribution on the surface area of the sample AC I(d), AC II(e), and AC III(f).

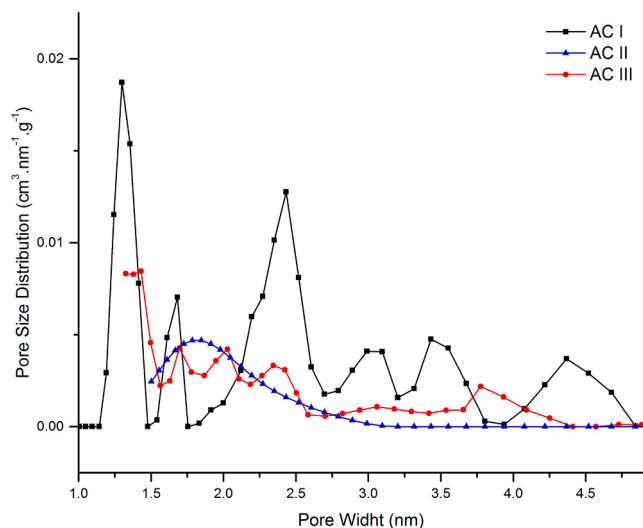


Fig. 6. The pore distribution of the activated carbon based-coconut shell by different chemical activator.

water filters, gas filters, heavy metal absorption, energy storage application, etc [52–54].

Table 4 compares the surface area characteristics of activated carbon based on a coconut shell and bio-waste reported by Kopac (2012). The characterization result of the samples shows that the AC I, which used NaOH activating agent, has the highest surface area, which is 516 m²g⁻¹. While the other sample in which AC II used ZnCl₂ and AC III used H₃PO₄ as an activating agent has a surface area of about 42 m²g⁻¹ and 23 m²g⁻¹, respectively. The difference in surface area from the samples can be caused by the chemical activator used as a representative of acid, salt,

and base properties [7].

The pore volume of micropore classification or pore size less than 2 nm shows the sampled AC I has 71.53 cm³g⁻¹. It is different from sample AC II and AC III, which have a small pore volume of 0.012 and 0.008 cm³g⁻¹, respectively. Meanwhile, the surface area of micropore classification on AC I, AC II, and AC III have shown 391.74 m²g⁻¹, 19.71 m²g⁻¹, and 17.97 m²g⁻¹, respectively. It can be seen that Fig. 5 (a,b,c) shows a different graphic pattern produced and were analyzed the pore volume of each sample, in which AC I, AC II, AC III have 0.160 cm³g⁻¹, 0.023 cm³g⁻¹, 0.011 cm³g⁻¹, respectively. Fig. 5 (d,e,f) shows the pore size distribution of the surface area which the data has a relationship with the pore size. The result of this characterization has confirmed that the number of pores formed in the micropore classification has a relationship with the surface area [13,37]. It can be seen that the surface area in the micropore classification corresponded to the micropore volume in each sample where a large pore volume, the surface area will also form dominantly in that classification [55]. Thus, the total surface area is produced in the mesoporous and macropore classification [50,56].

The pore size distribution of the sample is shown in Fig. 6. It shows that the pore size distribution of each sample was different due to the surface area and pore volume of the sample. Analysis result of pore size distribution produced a pore size of the sample where AC I, AC II, and AC III have a pore size about 1.29 nm, 1.43 nm, and 1.19 nm, respectively. And the average result of the diameter pore of AC I, AC II, and AC III has produced about 1.79 nm, 3.46 nm, and 4.48 nm.

Adsorption/desorption isotherms data were measured using nitrogen at 77 K are shown in Fig. 7 (a,b,c). The adsorption isotherm was used to evaluate the gas adsorbed at different relative pressures (P/Po), where P is the gas vapor pressure, and Po is the saturation pressure of the adsorbent. According to the data, the comparison of adsorption isotherms of three cases has shown the different volume values with quite a similar curve model. The curve of adsorption isotherm of coconut shell activated carbon produced using NaOH, ZnCl₂, H₃PO₄ chemical agents

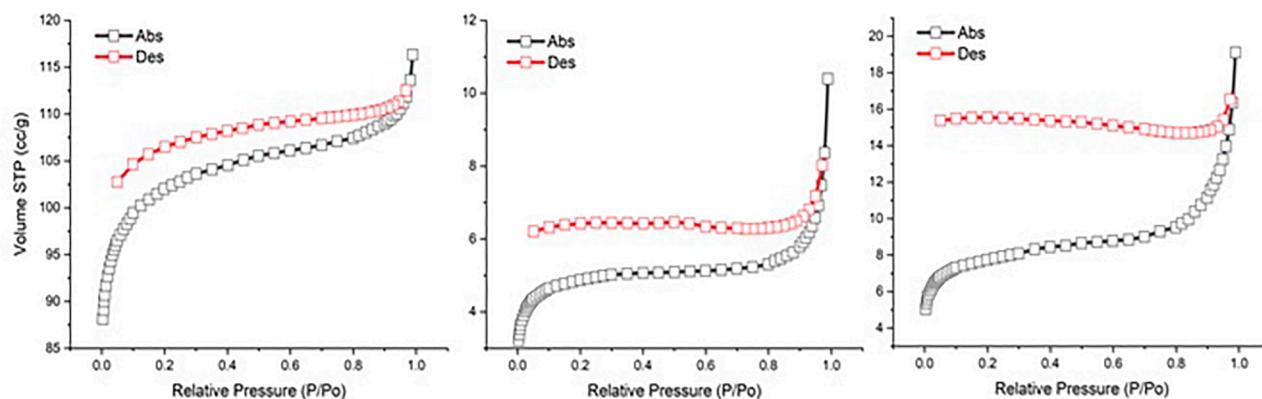


Fig. 7. Nitrogen adsorption-desorption of activated carbon based on coconut shell with different chemical agent : (a) NaOH, (b) ZnCl_2 , (c) H_3PO_4 .

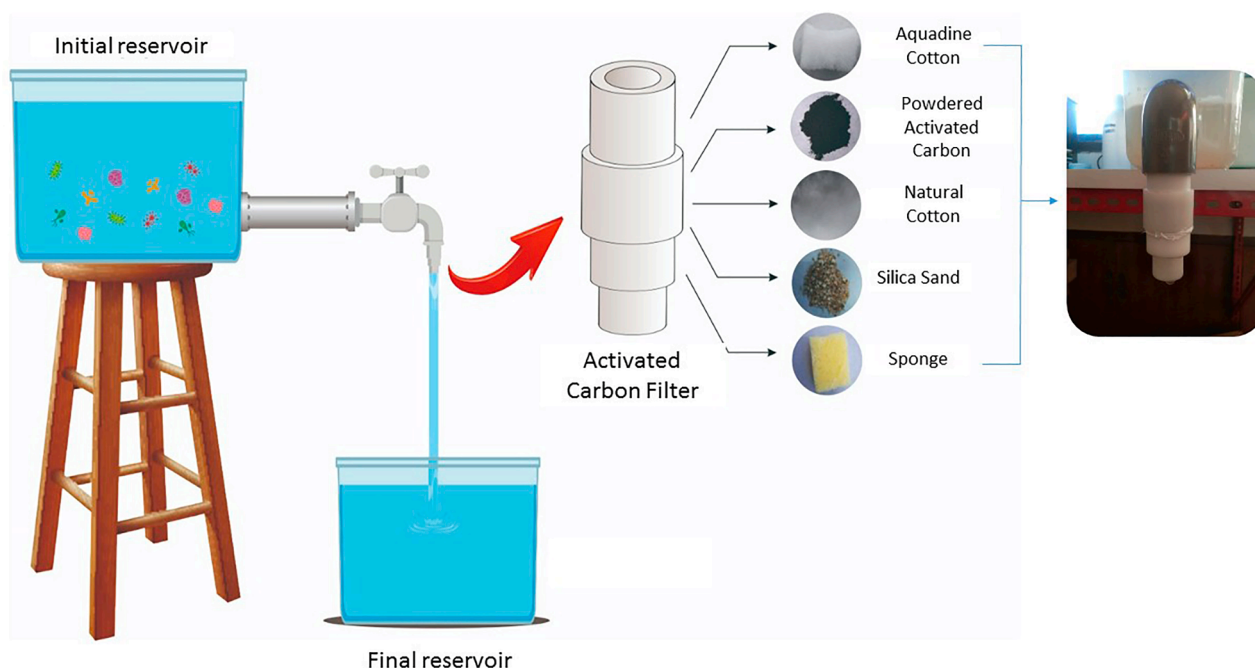


Fig. 8. Schematic of the polluted water filtration process.

suggests a Type II isotherm [57]. The data has shown a linear relationship between the adsorption and relative pressure at a moderate pressure ratio. At a higher-pressure ratio, the absorption quantity was slightly increased. The maximum amount adsorbed was 116.3 cc/g, 19.1 cc/g, and 10.3 cc/g at a relative pressure ratio of 0.9893, 0.98872, 0.9896p/po, respectively. A minimum adsorption quantity for AC I, II, and III was 88.12 cc/g, 5.04 cc/g, and 3.22 cc/g at approximately a pressure ratio of 0.0038.

TDS measurement

Total Dissolved Solid (TDS) measurements in polluted water were

conducted using the best activating agent, NaOH. Chemical activation using NaOH activator was carried out in a ratio of 1:1 (4 g charcoal : 4 g NaOH) mentioned as AC 1:1, 1:2 (4 g charcoal: 8 g NaOH) mentioned as AC 1:2, and 1:3, (4 g charcoal: 12 g NaOH) mentioned as AC 1:3, in which the three sample variants were dissolved with 10 ml of aquabides. Furthermore, the activation process was carried out. The three sample variants were dried in an oven to obtain a powder form, then used as the main material in the polluted water filtration process. Two different types of contaminated water were used, the first is well water, and the second is household wastewater. The scheme of the polluted water filtration process can be seen in Fig. 8.

After the filtration process was carried out and measured using a pH

Table 5

TDS and pH measurement of two variants polluted water before and after filtering by using activated carbon with NaOH activating agent.

Water Type	Water Before Filtering		Condition		Water After Filtering						Condition	
					Ac 1:1		AC 1:2		AC 1:3			
	pH	TDS	Color	Smell	pH	TDS	pH	TDS	pH	TDS	Color	Smell
Well Water	7.8	250	Dark Brown	No	7.2	250	7.2	193	7.2	248	Clear	No
Household Wastewater	7.9	296	Dark Black	Yes	7.1	250	7.2	187	7.3	293	Clear	No



Fig. 9. Illustration of well water and household wastewater before and after being filtered using activated carbon with an activating agent NaOH.

and TDS meter, the results were quite good, while the data can be seen in Table 5. For well water, water before being filtered has a characteristic dark brownish color but does not smell, and after being filtered, the color changes to clear. There was also a change in pH, where the pH before was 7.8, and the pH after being filtered decreased to 7.2. And for the TDS results before being filtered were 250 ppm, then after being filtered using three variants of activation agents AC 1:1, AC 1:2, AC 1:3 were 250 ppm, 193 ppm, 248 ppm, respectively. Furthermore, for household wastewater, the characteristics of water are cloudy blackish in color, and smelly, then after being filtered, it turns clear and odorless. For pH testing, the initial pH was 7.9, and the pH after filtering decreased to almost a neutral pH, which is 7.1, 7.2, and 7.3, respectively, for the three sample variants. Meanwhile, the TDS results before being filtered were 250 ppm, and after being filtered with AC 1:1, AC 1:2, AC 1:3, activation agents were 260 ppm, 187 ppm, and 293 ppm, respectively. The illustration of two types of wastewater before and after being filtered can be seen in Fig. 9.

According to the total dissolved solids, water quality is divided into three categories: TDS > 26–140 ppm of drinking water containing inorganic minerals, TDS > 140 ppm of ordinary drinking water, TDS > 500 ppm can be harmful to health [58,59]. Inorganic hazardous elements, organic chemicals, and microbes are the three types of water pollutants. Mercury, cadmium, lead, and chromium, copper, and so on are examples of metallic elements which could be found in the inorganic hazardous elements. Meanwhile, common organic pollutants in water include personal care items, pharmaceuticals, pesticides, endocrine disruptors, detergents, organic dyes, also industrial organic wastes such as halogens, aromatics, and phenolics [60]. Based on the data, water has been filtered using activated carbon with the activating agent of NaOH as a drinking water standard for the total dissolved solids in the water. Unfortunately, in this measurement, it is not possible to determine specifically the pollutants that have been filtered using activated carbon. Nevertheless the visible change in the color of the water and the pH value towards neutral indicates the contaminants in water have been successfully filtered.

Conclusions

In this research, high-quality microporous activated carbon has been successfully fabricated from coconut shell waste by using activating agents NaOH, ZnCl_2 , and H_3PO_4 . XRD results indicated that coconut shell charcoal has a high carbon content. FTIR characterization showed

that the oxygen functional groups in the sample were hydroxyl, epoxy, and carboxyl. The SEM results illustrated the formation of pores on the surface of the sample, which indicates the success of the activation process, also supported by SAA data which shows a reasonably good surface area, pore size, and pore volume. In addition, TDS measurements show that the role of activated carbon with the activating agent of NaOH has significantly improved the quality of polluted water. Therefore, the data results indicate that activated carbon based on coconut shell waste can be potentially applied as a wastewater treatment technology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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